

Advanced Reconfigurable Computing DLX Execute Requirements

Introduction

Your task is to implement and verify the EXECUTE stage of the DLX architecture in conjunction with your previously-implemented FETCH and DECODE stages.

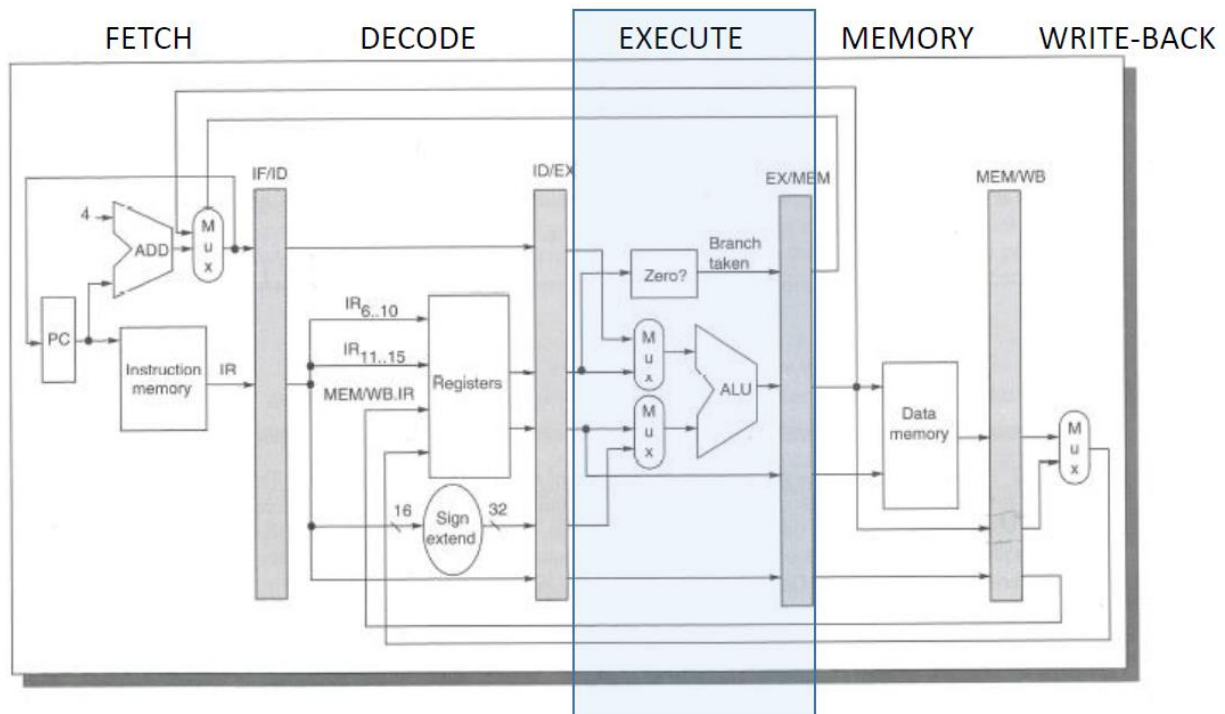


Figure 1: DLX 5-stage pipeline architecture with Execute stage highlighted

Requirements

1. The lab will be done in modular, hierarchical VHDL. Your top module should be named something like "DLX". Under this top level you should have a FETCH module, a DECODE module, and an EXECUTE module.
2. Implement the ALU, multiplexers, "Zero?", appropriate registers, and any needed control logic as shown in Figure 1, with the modifications discussed in class.
3. Include the previously-created VHDL package, modified if necessary, for defining constants and such that will need to be shared across the architecture.
4. Design appropriate Questa test-benches to exercise your FETCH, DECODE, and EXECUTE stages working together. Verify that your architecture successfully handles the four examples from the first lab (example1, example2, example3, and factorial). Exercise the four inputs to the system in a variety of ways.

Pass-off

10 pts – Demonstrate to the instructor a ModelSim simulation of your factorial example. Simulate branches, jumps and memory accesses in the appropriate spots in the code. Demonstrate that the FETCH, DECODE, and EXECUTE stages work correctly in the pipeline.